

LESSON 331a (1.5 GROUND)

Lesson Objective

The objective of this lesson is to evaluate the student's IFR skills in preparation for the End-of-Course Test.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

- Flight Operations Manual. (Version 6)
- Advisory Circulars. ([Search Tool](#))
 - Aviation Weather. ([AC 00-6B](#))
 - Aviation Weather Services. ([AC 00-45H](#))
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 07 - Aircraft Systems.
 - Chapter 08 - Flight Instruments.
 - Chapter 12 - Weather Theory.
 - Chapter 13 - Aviation Weather Services.
 - Chapter 15 - Airspace.
 - Chapter 16 - Navigation.
- Instrument Flying Handbook. ([FAA-H-8083-15B](#))
 - Chapter 05 - Flight Instruments.
 - Chapter 09 - Navigation Systems.
 - Chapter 10 - IFR Flight.
- Warrior III Pilot's Information Manual.
 - Section 01 - General.
 - Section 07 - Systems.
- Instrument Flying Handbook. ([FAA-H-8083-15B](#))
- Aeronautical Chart User's Guide. ([Guide](#))
- Terminal Procedures Publication. ([TPP](#)) - SE3.
- Aeronautical Information Manual.

Lesson Content

This lesson is to be used as a review of all of the line items in the instrument training course. There is a lot of information in this lesson plan. Only pick a few items from each line item to review with the student.

Pilot Qualifications

- Instrument Rating Privileges (61.3e):
 - What can the student do with their instrument rating?

- Fly in Class A
- Special VFR at night (sunset to sunrise)
 - Airplane must have the required instrument equipment and the pilot must have an instrument rating.
 - A VFR pilot is privileged to fly below VFR minimums with a VFR clearance.
 - 1SM and COC.
- File an IFR flight plan
- Fly in instrument meteorological conditions (IMC)
- IFR Flight Plan and Clearance Required:
 - When is an IFR flight plan required?
 - According to 14 CFR 91.173, no person may operate an aircraft under IFR in controlled airspace unless an IFR flight plan has been filed and a clearance has been received from ATC.
 - Is filing IFR enough to fly through IMC?
 - No, a clearance must be received from ATC.
 - Is it legal to fly in uncontrolled airspace in IMC without filing?
 - Although flying into IMC in uncontrolled airspace without a clearance is technically legal, it has been found in the past to be a violation of **91.13 - Careless or reckless operation**.
- Currency as an Instrument Pilot:
 - How long do instrument privileges last after the practical test?
 - After the initial 6 months have passed, you have an additional 6 months to regain currency with a safety pilot, instructor, examiner, OR in the simulator (grace period).
 - What must be done to retain currency?
 - Within the 6 preceding calendar months (ATD or Airplane):
 - 6 approaches.
 - Holding procedures (minimum of 1 hold)
 - Intercepting and tracking courses.
- Safety Pilots (91.109):
 - When must a safety pilot be used?
 - A safety pilot must be used during any flight in an aircraft during the grace period.
 - Who can be a safety pilot?
 - A safety pilot must hold a minimum of a PPL in the same category, class, and type (if required) of the aircraft being flown.
 - Can a Private Pilot be a safety pilot in an Arrow without a complex endorsement?
 - Yes, a safety pilot is not required to have the endorsements required for the specific aircraft of use.
 - Can a Private Pilot be a safety pilot in a Seminole without AMEL?
 - No, because the safety pilot does not have a PPL in the same class (MEL).

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- Does the safety pilot need to be current?
 - No, but they cannot enter IMC or act as PIC unless they are current.
- What type of flight controls must the airplane possess to use a safety pilot?
 - Airplane (single-engine only) must have dual controls or is equipped with a single throw-over control wheel.
- What if the safety pilot cannot see well enough?
 - The Safety pilot must have adequate vision forward and side to side.
 - If not, a competent observer may substitute.
- What kind of time may be logged by the safety pilot? The instrument pilot?
- Conditions Required to Log Tasks for Currency (61.57):
 - What conditions are required to log an approach?
 1. You must operate the aircraft, FFS, ATD or FTD solely by reference to instruments.
 2. Be established on each required segment of the IAP down to its published minimums (MDA or DA).
 3. The Initial (starting at the IAF), Intermediate and Final approach segments are required for logging the approach, unless:
 - When radar vectored to final by ATC. (In simulated IMC, vectors from the safety pilot are also accepted).
 4. If conducted in simulated IMC in an aircraft, or in a FFS, ATD or FTD, simulated conditions must continue down to MDA or DA.
 5. In an aircraft, flight must be in actual or simulated IMC.
 6. In an aircraft, if conditions change from actual IMC to VMC during the final approach segment, you can still log the approach.
 - Define actual instrument conditions.
 - IMC: conditions that require PIC to utilize sole reference to the instruments to maintain the desired route of flight.
 - IFR: conditions that the instrument rules apply (<3sm and/or <1,000ft ceilings).
 - Can a safety pilot be brought into IMC if the pilot flying is not current?
 - Yes, as long as they are instrument rated and current.
- Instrument Proficiency Check (61.57):
 - When are IPCs required?
 - After the 6-month grace period has ended.
 - Who may administer an IPC?
 - Conducted by a DPE, US Armed Forces (provided the IPC is for military personnel), company check pilot (Part 121, 125, and 135), CFII, or anyone authorized by the FAA.
 - What tasks are involved in an IPC?
 - (i) Air traffic control clearances and procedures;
 - (ii) Flight by reference to instruments;

- (iii) Navigation systems;
- (iv) Instrument approach procedures;
- (v) Emergency operations; and
- (vi) Postflight procedures.
- Can an IPC be administered in a simulator?
 - AATDs can be utilized for the majority of the IPC as specified in the Letter of Authorization issued for the device. However, the circling approach, the landing Task, and the multiengine airplane Tasks must be accomplished in an aircraft or FFS (Level B, C, or D). A BATD cannot be used for any part of the IPC.

Weather Information

- What weather sources fulfill the legal preflight requirements?
 - Calling a weather briefer.
 - 1800wxbrief.com.
 - ForeFlight generated weather briefing.
 - Aviation Weather Center.
- Acceptable Weather Products and Resources.
 - METAR:
 - What is a METAR?
 - Observation of current surface weather.
 - How long is a METAR valid for?
 - Valid for one hour.
 - How often is a METAR issued?
 - Issued hourly unless significant changes have occurred.
 - What is a SPECI?
 - A special METAR (SPECI) can be issued between hourly reports (like wind direction changing by 45 degrees or more in less than 15 minutes, more in AC).
 - TAF:
 - What is a TAF?
 - Terminal aerodrome forecast
 - How long is a TAF valid for?
 - Valid for 24 up to 30 hours.
 - How often is a TAF issued?
 - Updated 4 times a day.
 - Winds/Temps Data:
 - What are FB Winds?
 - A summary of winds aloft forecast for the route/area as interpolated from forecast data for the local and/or the adjacent reporting locations for level through 12,000 feet.
 - How often are FB Winds issued?

Instrument Rating Course

- On a global scale, issued 4 times a day (Every 6hrs).
- How far out does each issuance forecast?
 - Forecasts out to 6, 12 & 24hrs.
- PIREP:
 - What is a PIREP?
 - Pilot weather reports.
 - Weather report from a flight deck. It consists of message type, location, time, altitude/flight level, type of aircraft, and at least one other element to describe the report phenomena.
 - How often are they issued?
 - Ceilings at or below 5,000 feet.
 - Visibility reported on the surface or aloft is 5 miles or less.
 - Thunderstorms and related phenomena.
 - Turbulence of moderate degree or greater.
 - Icing of light degree or greater.
 - Wind shear.
 - Braking action reports less than good.
 - Volcanic eruption, ash clouds, and/or detection of sulfur gasses in the cabin.
- Prognostic charts:
 - Low-Level Significant Weather Chart (Surface to FL240):
 - Quiz the student on the information shown on a prognostic chart.
 - Surface Prognostics Chart:
 - Quiz the student on the information shown on a surface analysis chart.
- AIRMETs:
 - What is an AIRMET?
 - AIRMET (WA) – Advises of light to moderate weather that may be hazardous to aircraft (primarily light).
 - What is the difference between an AIRMET and a G-AIRMET?
 - G-AIRMET is a snapshot or a graphical depiction of a hazard at that given time using lat/long coordinates, whereas an AIRMET is a textual description using VORs to describe the affected area.
 - How often are they issued, and how long are they valid for?
 - G-AIRMETs are issued every 3 hours, whereas traditional AIRMETs are issued every 6 hours. AIRMETs are generated by the software using the initial, 3hr & 6hr forecast.
 - What are the different types of G-AIRMETs?
 - Sierra:
 - IFR conditions (ceilings less than 1000 ft, visibility less than 3sm).
 - Mountain obscurations.
 - Tango:

Instrument Rating Course

- Light to moderate Turbulence.
 - Surface winds 30 knots or greater.
 - Low-level wind shear.
- Zulu:
 - Icing.
 - Freezing levels.
- What condition must be met for an G-AIRMET to be issued?
 - A G-AIRMET will be issued if any of these conditions are expected to happen over an area of at least 3,000 square miles.
- SIGMETs:
 - What is a SIGMET?
 - SIGMET (WS) – Advises of severe weather conditions that may be hazardous to all aircraft.
 - How often are they issued, and how long are they valid for?
 - Unscheduled (comes out when needed).
 - Valid for 4 hours (6 for hurricanes).
 - What is a SIGMET issued for?
 - Severe icing not associated with thunderstorms.
 - Severe or extreme turbulence or CAT not associated with thunderstorms.
 - Dust/sandstorms.
 - Volcanic ash.
 - What condition must be met for a SIGMET to be issued?
 - Issued when these weather conditions are forecasted to happen over an area of at least 3000 square miles or an area judged to have a significant impact on aircraft safety.
- Convective SIGMETs:
 - What is a convective SIGMET?
 - Convective SIGMET (WST) – Hazardous convective weather affecting every flight.
 - Only issued for the contiguous United States.
 - How often are they issued, and how long are they valid for?
 - Issued at 55 minutes past the hour.
 - Valid for 2 hours.
 - What conditions must be met for a convective SIGMET to be issued?
 - Severe thunderstorms with surface winds greater than 50 knots
 - Hail at the surface at least 3/4 inch in diameter
 - Affects 40% or more of a 3000 sq miles area.
 - Line of thunderstorms at least 60 miles long with thunderstorms affecting at least 40 percent of its length.
 - Tornadoes

- Convective Outlook:
 - What is a convective outlook?
 - Consists of both a narrative and a graphic showing severe thunderstorm threats across the US.
 - How often are they issued, and how long are they valid for?
 - Day 1: risk of severe weather for 24 hours. Issued 5 times a day.
 - Day 2: risk of severe weather for the next 24 hours. Issued 2 times a day.
 - Day 3: risk of severe weather for the next 24 hours. Issued daily.
 - Day 4-8: risk of severe weather for day 4 through day 8. Issued daily.
- Center Weather Advisory:
 - What is a center weather advisory?
 - A CWA is an aviation weather warning for conditions meeting or approaching national in-flight advisory (AIRMET, SIGMET, or Convective SIGMET) criteria.
 - Who issues them?
 - CWA are issued by the NWS CWSU (Center Weather Service Unit). The 2 centers in Florida are Jacksonville & Miami.
 - How long are they valid for?
 - CWAs are valid for up to 2 hours and may include forecasts of conditions expected to begin within 2 hours of issuance.
- Meteorology:
 - Atmospheric composition and stability:
 - What is atmospheric stability?
 - The stability of the atmosphere depends on its ability to resist vertical motion.
 - Air masses and fronts:
 - What is an air mass?
 - A body of air with equal or uniform properties (humidity, temperature, pressure, etc.)
 - What is a cold front? What are the characteristics of a cold front?
 - Occurs when a mass of cold, dense and stable air advances and replaces a body of warmer air.
 - Moves at 25-30 mph, can move as fast as 60 mph.
 - Rapidly ascending air causes the temperature to decrease suddenly, forcing the creation of clouds.
 - It “lifts moist air and creates instability” - a recipe for thunderstorms.
 - Before the cold front passes, cirriform towering cumulus clouds are present, and cumulonimbus clouds may develop.
 - Showery precipitation.

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- After frontal passage, clouds dissipate to cumulus clouds with a corresponding decrease in precipitation
- Good visibility prevails and winds from the west and northwest
- What would a pilot expect to encounter as they fly through a cold front?
 - Flight progresses to the oncoming cold front, the clouds show signs of vertical development with a broken layer.
 - Conditions may become severe with strong storms.
 - Conditions will improve rapidly.
 - Pilots should use sound judgment on frontal passage.
- What is a warm front? What are the characteristics of a warm front?
 - Occurs when the warm mass of air advances and replaces a body of colder air. Warm fronts move slowly, 10-25 mph. The slope of the advancing front slides over the top of the cooler air and gradually pushes it out of the area. As the warm air is lifted, the temperature drops and condensation occurs from the humidity of the air.
 - Generally cirriform, stratiform clouds (stable air) along with fog can be expected to form along the frontal boundary (rain evaporates when it gets close to the cooler ground).
 - Light to moderate steady precipitation.
- What would a pilot expect to encounter as they fly through a warm front?
 - Visibility decreases.
 - Falling barometric pressure.
 - Weather overall deteriorates.
 - May be impossible to continue a VFR flight after flying through a warm front.
- What is a stationary front? What are the characteristics of a stationary front?
 - When the forces of two air masses are relatively equal, the boundary layer or front that separates them remains stationary and influences the local weather for days.
- What is an occluded front? What are the characteristics of an occluded front?
 - Fast moving cold front catches up with a slow-moving warm front. Warm front weather prevails but is followed by cold front weather.
- What is frontogenesis? What about frontolysis?
 - Frontogenesis: the generation of a front.
 - Frontolysis: the dissipation of a front.
- Wind:
 - How does air flow relative to the isobars at the surface? And aloft?
 - Winds flow across isobars at the surface due to friction and reduction of Coriolis force.
 - Winds flow parallel to the isobars at altitude (Winds aloft).
 - What does a tight pressure gradient indicate in terms of wind?

Instrument Rating Course

- Close isobars on charts indicate a strong pressure gradient = change in winds.
- Temperature:
 - What is the temperature lapse rate on a standard day?
 - -2°C/1,000ft of altitude gain.
 - What is a temperature inversion?
 - The temperature increases with altitude.
 - When might a temperature inversion be experienced?
 - Surface-based temperature inversions occur on clear, cool nights when the air close to the ground is cooled by the lowering temperature of the ground.
- Weather system formation, including air masses and fronts:
 - What is the source of all weather?
 - Uneven heating of the earth's surface.
- Clouds:
 - What are some types of clouds?
 - High clouds [Cirrus (Ci), cirrocumulus (Cc), and cirrostratus (Cs)]
 - Medium clouds [Altostratus (As), altostratus (As), and nimbostratus (Ns)]
 - Low clouds: [Cumulus (Cu), towering cumulus (TCu), stratocumulus (Sc), stratus (St), and cumulonimbus (Cb)]
 - What counts as a ceiling?
 - Broken (BKN) or Overcast (OVC).
- Turbulence:
 - What is the difference between turbulence and chop?
 - Turbulence is a random, sporadic movement of the aircraft.
 - Chop is a rapid and somewhat rhythmic movement of the aircraft.
 - What are the intensities of turbulence or chop?
 - Light
 - Moderate
 - Severe
 - Extreme
 - What are the frequencies of turbulence?
 - Occasional
 - Intermittent
 - Continuous
- Thunderstorms and Microbursts:
 - What are the criteria required for the formation of a thunderstorm?
 - Lifting action.
 - High moisture.
 - Unstable air.
 - What are the stages of a thunderstorm?

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- Developing stage:
 - Towering cumulus (rising air), little rain, occasional lightning.
- Mature:
 - Fully formed, heavy rain, frequent lightning, strong winds. All hazards are at their worst; therefore this stage is considered to be the most dangerous.
- Dissipating:
 - Downdrafts, rainfall decreases, still producing strong winds and lightning.
- What are the different types of thunderstorms?
 - Single cell: develop on warm, humid days. Severe, produce hail, and microbursts.
 - Usually associated with daytime convection.
 - Also referred to as air mass.
 - Multi-cell: develop in clusters, cover large areas, and can move in different directions.
 - Usually associated with large-scale weather systems such as low-pressure systems and fronts.
 - Also referred to as steady state.
 - Squall line: a narrow band of thunderstorms that form ahead of a cold front (moist, unstable air, lifting action). Severe and wide. Can move very quickly.
 - Supercell: Single long-lived thunderstorm. Large hail and significant tornadoes.
- What weather can we expect in a thunderstorm?
 - Lightning.
 - Turbulence.
 - Hail.
 - Tornadoes.
 - Wind shear.
 - Icing.
- Icing and freezing level information:
 - What is the difference between known icing and known icing conditions?
 - Known icing: visible accretion (PIREPs).
 - Known icing conditions: forecasts and reports (airmets, freezing level charts, winds and temps aloft, etc.).
 - What are the performance impacts of icing?
 - Drag → increase.
 - Lift → decrease.
 - Weight → increase.
 - Stall speed → increase.

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- What are the different categories of icing in general?
 - Structural
 - Requires visible moisture and aircraft surface below freezing.
 - Clear → forms when large supercooled drop hits the aircraft and spreads and freezes slowly.
 - Cumulus clouds.
 - Clear in color.
 - Hits leading edge and freezes as it reaches trailing edge.
 - -10 to 0 Celsius.
 - Commonly found in cold fronts.
 - Rime → forms when small drops hit the aircraft and freeze rapidly.
 - White color.
 - Stratus clouds.
 - Freezes instantly.
 - -20 to -10 Celsius.
 - Commonly found in warm fronts.
 - Mix → mix of the two.
 - Found in thunderstorms where there is a mix of stratus and cumulus clouds.
 - It is also found in stationary or occluded fronts.
 - Induction
 - Carb ice → can happen with temperatures as high as 100F/38C and a relative humidity of 80%.
 - Air in the carburetor is accelerated in the venturi, causing the temperature of the air to decrease. More susceptible to icing.
 - Throttle Ice → ice build-up in venturi due to the temperature of the air decreasing (faster air = colder air).
 - Fuel vaporization ice → cooling effect when fuel mixes with air and vaporizes in the carburetor. Moisture condenses out of the air, and water starts freezing in the carburetor.
 - Impact ice → As cold, moist air contacts anything solid, ice can start to form. This can happen on the air intake, filter, or carburetor.
 - What can the pilot do when dealing with induction icing?
 - Turn carb heat on (small reduction in RPM, warm air is less dense than cold air, enriches the mixture).
 - Instrument
 - Icing tends to form on small exposed surfaces first:
 - Pitot tube and static ports.

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- Static wicks.
 - Antennas.
 - Stall warning tab.
 - OAT probe.
- What are the rates of icing accretion?
 - Trace
 - Light
 - Moderate
 - Severe
- What is the difference between an anti-ice and de-ice device?
 - Anti-Icing includes preventative measures.
 - Deicing equipment removes ice once it has occurred.
- Fog/mist:
 - How does fog form?
 - The temperature of air near the ground is cooled to the air's dew point.
 - What are the different types of fog?
 - Radiation fog – Occurs when the ground cools rapidly due to terrestrial radiation, and surrounding air reaches the dew point. Does not form on cloudy nights because clouds trap heat. As the sun rises and the temperature increases, the fog burns off.
 - Advection fog – when a layer of warm moist air moves over cold land/water. 15-knot wind is necessary. Common to coastal areas where sea breezes can blow the air over cooler landmasses.
 - Upslope fog – Occurs when moist, stable air is forced up sloping land features like a mountain range. Air is colder at higher elevations which cools air to dewpoint. Also requires wind for the formation and continued existence.
 - Steam fog – or sea smoke, forms when cold dry air moves over warm water. As water evaporates it turns into smoke.
- Frost:
 - What is frost?
 - Ice crystal deposits are formed when temp and dewpoint are below freezing.
- Wind shear:
 - What is wind shear?
 - A dramatic shift in wind speed and/or direction.
 - What altitude does it occur at?
 - Wind shear can occur at any altitude.
- Microburst:
 - What is a microburst?
 - A strong downdraft that induces an outburst of damaging, divergent winds as high as 75 m/s on or near the ground over an area of 1-4 km.

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- What conditions would a pilot experience while flying through a microburst?
 - The plane will encounter an increase in headwind, then performance decreasing downdrafts, then a shear tailwind, and can result in terrain impact.
- Sea breeze and land breeze
 - How does the sea breeze form?
 - During the day - land warms faster than water – air over land becomes warmer and less dense. It rises and is replaced by cooler, denser air flowing over the water.
 - How does the land breeze form?
 - At night - land cools faster than water, as does corresponding air. Warmer air over the water rises and is replaced by cooler, dense air from the land.

Aircraft Systems and Aircraft Flight Instruments Related to IFR Operations

- Required Equipment - 91.205 (d)(e), 91.131(c)(1), 91.215, 91.411 and 91.413.
 - What equipment is required to fly IFR (day and night)?
 - VFR Day (ATOMATOFLAMES):
 - Airspeed Indicator.
 - Tachometer (for each engine).
 - Oil Pressure gauge.
 - Manifold Pressure (for constant speed props, supercharged, turbocharged).
 - Altimeter.
 - Temperature Gauge (liquid-cooled engine).
 - Oil temperature gauge.
 - Fuel Quantity Gauge.
 - Landing Gear Position Lights (if retractable).
 - Anti-Collision Lights (aircraft certified after March 11th, 1996).
 - Magnetic direction indicator.
 - ELT.
 - Seatbelts.
 - VFR Night (FLAPS):
 - Fuses/resettable circuit breakers.
 - Landing Light (Electric) for hire.
 - Anti-Collision Lights.
 - Aircraft certified after August 11th, 1971.
 - Aircraft certified before that date STILL NEED anti-collision lights, but the standards/requirements are different.
 - Position Lights.
 - Source of Electricity (Alternator/Battery).
 - IFR (GRABCARD):

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- Generator or alternator.
- Rate of turn indicator.
- Attitude indicator.
- Ball (slip/skid indicator).
- Clock (digital or analog with sweeping second hand)
 - Has to be installed in the aircraft. It can be a part of another system (ex: MFD's clock).
- Altimeter (sensitive altimeter).
- Radios/nav equipment appropriate for the flight.
- Directional gyro.
- DME (only required above FL240).
- VOR, TACAN, GPS, in Class B.
- Ask the student various questions regarding the pieces of equipment that they listed (e.g, manifold pressure gauge, temperature gauge, alternator vs. generator, source of power, etc.)
- Pitot-Static System:
 - Ask the student to explain/draw the pitot-static system.
 - Airspeed indicator:
 - How does the airspeed indicator work?
 - Pitot tube connected to DIAPHRAGM.
 - Static port connected to CASING.
 - Diaphragm is like a balloon that expands as more air enters it. The expansion is read as an increase in airspeed. Resisting the expansion is the static air in the casing. The difference between static and dynamic is pressure differential.
 - Differential pressure (airspeed read) = static + ram (pitot).
 - What if, during a climb, airspeed increases?
 - The pitot and drain hole are blocked.
 - What if, during a descent, airspeed increases?
 - The static system is blocked.
 - Ask what to do if there is a blockage.
 - Pitot heat
 - Alternate static source
 - Prior to operation:
 - Close all vents in the cockpit and storm window.
 - Turn on the heater and defroster.
 - Instrument indications:
 - Airspeed indicator reads higher than normal.
 - Altimeter reads higher than normal.
 - VSI shows a momentary climb.

- Lower pressure in the cabin due to leakage.
- What are the types of airspeed?
 - Indicated airspeed - Read directly from the indicator.
 - Calibrated airspeed - Indicated airspeed corrected for installation/position error of pitot mast.
 - Equivalent airspeed - Calibrated airspeed corrected for compressible airflow error (>10,000 MSL and >250 kts).
 - True airspeed - Equivalent airspeed corrected for non-std temp and press.
 - Groundspeed - True corrected for wind.
 - Speed of movement across the ground – used for ETE).
 - Mach # - Ratio of true airspeed to the local speed of sound.
- Altimeter:
 - How does the altimeter work?
 - Casing is attached to the static port.
 - Aneroid Wafer is in casing sealed to 29.92in Hg (3 for redundancy).
 - Aneroid Wafer expands or contracts against static pressure in the casing, reading an increase or decrease in altitude.
 - In a climb, there is a decrease in static pressure, so the static pressure in the case exits, allowing the aneroid wafer to expand, showing an increase in altitude.
 - The opposite is true for a descent.
 - Gears and linkages connected to needles.
 - Pressure can be adjusted through the Kohlsman window → sensitive altimeter.
 - 1 in hg is equal to 1000 ft.
 - What are the errors?
 - Flying from high pressure to low pressure (without adjusting the altimeter setting): Airplane will be lower than indicated.
 - This is because low pressure enters the altimeter case, so the aneroid wafers expand and indicate that the plane is higher. As a result, the pilot pitches down to return to the assigned altitude, making the plane lower than indicated.
 - Flying from hot to cold weather (without adjusting the altimeter setting): Airplane will be lower than indicated.
 - What are the types of altitudes?
 - Indicated altitude - Read off the altimeter.
 - Pressure altitude - Height above the standard datum plane.
 - Pressure altitude = Indicated altitude when Kohlsman set to 29.92" Hg
 - $PA = (Std. - \text{current altimeter setting}) \times 1000 + \text{field elevation}$.
 - Density altitude - Pressure corrected for non-std temp.

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- $DA = PA + (120 \times (OAT\ ^\circ C - ISA\ temperature\ ^\circ C))$.
- Higher DA = Worse performance.
 - Reduced Engine Output.
 - Reduced propeller efficiency.
 - Reduced Flight control effectiveness.
- True altitude - Height above sea level (MSL).
 - True altitude = Indicated altitude when Kohlsman is set to the current pressure setting.
- Absolute altitude - Height above the ground or terrain (AGL).
- Vertical speed indicator:
 - How does the vertical speed indicator work?
 - Static port is connected to BOTH the CASING and DIAPHRAGM.
 - Calibrated leak (6-9 second lag) in the CASING causes static air to **leave** the casing **SLOWER** than the diaphragm leading to a pressure differential read as a vertical speed.
 - Measures rate of climb/descent.
 - Pressures are equalized when level, thus showing 0ft up or down.
 - Descent: diaphragm expands.
 - Climb: diaphragm contracts.
 - Is the aircraft IFR legal if the VSI reads +200 fpm on the ramp?
 - Yes, but 200fpm is the new zero point.
- Vacuum system:
 - Ask the student to explain/draw the system.
 - Air enters the vacuum system through a filter, passes through the air-driven gyro instruments and the suction gauge, into the vacuum regulator, and on to the vacuum pump from which it is expelled through a short pipe.
 - What does the vacuum air do to the instruments?
 - Vacuum air is used to spin the gyroscopes in the gyroscopic instruments.
 - What if the system fails? What is the indication?
 - There is a shear drive that protects the engine accessory drive from damage due to a seized pump. The shear drive will disconnect the pump from the engine.
 - Low suction due to:
 - Faulty vacuum regulator.
 - Dirty filter.
 - System leak.
 - Low engine RPM.
 - No suction due:
 - Sheared pump drive.
 - Defective pump.
 - Defective gauge.

- Collapsed line.
 - Do we have a vacuum system on our airplane?
 - We do NOT have a vacuum system in the Warrior Avidyne.
- Gyroscopic instruments:
 - What are the gyroscopic principles?
 - Rigidity in Space:
 - Ability of gyro to remain in a fixed position in the plane in which it is spinning.
 - Gimbals allow the aircraft to rotate freely in any direction and the gyro to remain fixed.
 - Precession:
 - Force felt 90 degrees in the same plane of rotation.
 - Attitude indicator:
 - What is the operating principle?
 - Rigidity in space.
 - How does it work?
 - Horizontally mounted gyro that rotates on the vertical axis with dual gimbals.
 - The front-facing horizon card is connected to the horizontal gimbal.
 - The airplane rotates about the gimbals.
 - Gyro rotates counterclockwise (18000-21000 RPM)
 - 4 air-ports covered by pendulous vanes use precession to stabilize the gyro (one port opens more than the other to reset the gyro upright).
 - This is used in case the gyro isn't upright.
 - Rotor tilts, but pendulous vanes remain vertical (gravity), exhaust port opens fully on one side, closes on other side, causing increased force from the open port. Precession causes force to act 90 degrees to that, erecting the gyro.
 - What is the power source for the instrument in our airplane?
 - Our Warrior Avidyne uses a standby attitude indicator whose gyroscope is battery powered.
 - If the training airplane's attitude indicator is electrical, when does the battery come on?
 - Upon loss of aircraft power.
 - How long does this battery last?
 - 1 hour.
 - What errors are associated with the attitude indicator?
 - Accelerating causes a false climb indication.
 - Decelerating causes a false descent indication.
 - Skid: shows turn in the opposite direction of skid.
 - What are the limitations of the attitude indicator?
 - Banking more than 110 degrees.

- Pitching more than 70 degrees.
- Heading indicator:
 - What is the operating principle?
 - Rigidity in space.
 - How does it work?
 - Vertically mounted gyro that rotates on the horizontal axis.
 - Spins away from the pilot.
 - Gyro speed: 10,000 - 18,000 RPM
 - Movement of the yaw gimbal transmitted to the instrument through gears.
 - Air nozzles: Two air jets will shoot air into the gyro if it precesses. They shoot air on one side to bring it back upright.
 - What errors are associated with the heading indicator?
 - Error about 3 degrees every 15 minutes due to precession & friction.
 - Realign every 15 minutes with the mag compass.
 - What are the limitations of the heading indicator?
 - Exceeding 55 degrees of bank or pitch may cause the gyro to tumble (spin erratically) and become inaccurate.
- Turn coordinator:
 - What is the operating principle?
 - Precession.
 - What is its power source?
 - Our turn coordinator is electrically powered by the aircraft's electrical system.
 - How is this instrument different from a turn and slip indicator?
 - Turn coordinator has a gyro that is canted back 30 degrees and a set of calibrated springs that allow it to sense rate-of-roll.
 - Rate of turn: the number of degrees of heading change per unit of time.
 - Rate of roll: measures the speed at which an airplane rotates around its longitudinal axis.
 - What is the failed indication?
 - A red flag on the face of the instrument.
- Electrical system:
 - Ask the student to describe the batteries, alternators, including backups in terms of amps, amp-hours, and volts.
 - Battery:
 - 24 volts and 35 amp-hours.
 - Used for engine starting: powers the starter motor and gives initial charge to the alternator.
 - Used as a backup in case the alternator and standby alternator fail.
 - Alternator:

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- 28 volts and 70 amps.
 - Driven by the alternator belt connected to the engine.
 - The alternator belt spins the pulley, which spins the rotor (rotating electromagnet) inside the stator (copper windings).
 - Standby alternator:
 - 28 volts and 20 amps (2500 RPM).
 - Activates if bus voltage drops to 26 volts.
 - Have the student explain how electricity flows from the battery to the main bus.
 - How is the electrical system protected?
 - Voltage regulator:
 - Attempts to hold a steady electrical output (28V to 30.5).
 - Overvoltage relay:
 - Takes alternator offline when the voltage regulator isn't properly regulating the voltage.
- Autopilot:
 - What is the model name of the autopilot?
 - S-TEC 55X
 - How does the autopilot work?
 - The autopilot controls movement on two axes, longitudinal and lateral. Attached to the cables and pulleys which move the control surfaces are 3 servos that act as actuators to increase or decrease tension based on commands sent from the autopilot.
 - What are the servos, and where are they?
 - Roll servo (located under the cabin floor)- Linked to turn coordinator.
 - Pitch servo (located under the baggage compartment floor)- Receives data from PFD, altimeter, VS, and pressure altitude transducer. Static pressure is received from the port located on both sides of the fuselage.
 - Trim servo (located behind the baggage compartment)- Electric trim must be on.
 - How can the autopilot be disconnected?
 - Press the autopilot disconnect button on the yoke, turn off the autopilot switch on the dashboard, or overpower the servos using the yoke!
 - What is the difference between the flight director and autopilot?
 - The flight director shows where to position the aircraft on the PFD to follow a flight track.
 - The autopilot uses the servos to actually fly this track.
 - When is autopilot use permitted?
 - The autopilot may be used at any time at or above 1,000 ft AGL and during the descent to the MDA or DA.
 - What does Avidyne recommend if the PFD fails in IMC?
 - Engage the autopilot.

- Avidyne system:
 - What are the components of the PFD?
 - Air data computer (ADC):
 - Located on the back end of the PFD unit.
 - Provides digital airspeed, altitude, VS, and outside air temperature (OAT) information and computed winds aloft and true airspeed (TAS) to the PFD for display.
 - ADC is fed information from the pitot mast.
 - Ram and static lines are piped directly to the ADC.
 - The pressure transducers are in the ADC.
 - The Outside Air Temperature Probes are used to sense and send OAT to the air data computer (ADC) for processing.
 - Located under the left wing.
 - Another one is located in the MFD and monitors system temperature too.
 - The ADC gathers this information and uses it to compute pitot/static data and translate it to readable data on the PFD.
 - If there is an ADC error, red X's will display where ADC instruments are.
 - Pitot static blockages DO NOT mean red X's.
 - Attitude and heading reference system (AHRS):
 - Located in the back of PFD.
 - Provides attitude, heading, and turn rate data.
 - Receives information from the:
 - Magnetometer (or flux gate):
 - Senses the Earth's magnetic fields and electrically transmits that information to the HSI.
 - HSI combines a heading indicator, omni-bearing selector, and bearing pointer.
 - HSI does not precess as there is no gyroscope.
 - Located on the left wing to be as far away from magnetic disturbances.
 - Accelerometers:
 - Takes measurements from 3-axis solid-state inertial sensors similar to tuning forks (pitch, roll, yaw).
 - With 3 of these tuning forks mounted on three different axes (90 degrees from each other), any change in acceleration (plane) is detected as a change in voltage.
 - Each tuning fork vibrates at a specific frequency when the aircraft's attitude is changed. It changes the vibration frequency

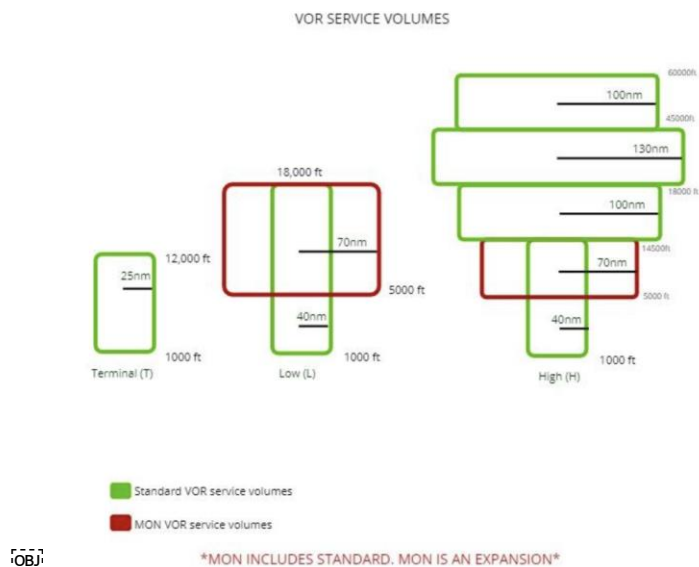
and, in result, changes the voltage that is then translated to the PFD as an attitude.

- What are the limitations of the PFD?
 - IFR flight is prohibited when the PFD or any standby instrument is inoperative.
 - IFR flight is prohibited upon aircraft total loss of essential engine parameter display.
 - The Avidyne FlightMax Entegra series Primary Flight Display Pilot's Guide must be available to the pilot during all flight operations.
 - If VLOC is displayed on the HSI and GPSS mode is engaged on the autopilot, the autopilot will track the active flight plan in the GPS corresponding to the selected VLOC.
 - GPSS mode must not be used on the final approach segment of a VLOC approach.
- What are the components of the MFD?
 - Data acquisition unit (DAU):
 - DAU receives information from pressure transducers and sensors throughout the engine (EIS - Engine Indicating System).
 - MFD can show all of the engine information.
 - It has a temperature probe to monitor system temperatures.
 - In case of a DAU failure, there will be red Xs over the engine indications.
- What are the limitations of the MFD?
 - The moving map display must not be used as the primary navigation instrument.
 - Use of the map page during IFR flight requires an IFR-approved GPS receiver and installation.
 - The Avidyne FlightMax EX5000 Multi-function Display Pilot's Guide must be available to the pilot during all flight operations.
 - Aircraft dispatch is prohibited when the MFD is inoperative.
 - Lightning overlay - N/A to our aircraft

Navigation Equipment

- VOR:
 - How does the VOR work?
 - 2 components: ground based VOR and your aircraft's receiver.
 - Ground VOR Emits two signals: variable and reference.
 - Variable Phase: VOR emits a signal per degree 30 times a second (kind of like a lighthouse).
 - Reference Phase: Sends out a 360 degrees pulse or broadcast of all of the radials.
 - The difference between the reference and variable phase is the phase shift and is calculated by your airborne VOR receiver to determine the radial you are on.

- What is the frequency range for VORs?
 - 108.0 - 117.95 MHz - identified with a morse code identifier.
- What are the different types of VORs?
 - VOR – The VOR by itself provides magnetic bearing information to and from the station.
 - VOR/DME – When DME (Distance Measuring Equipment) is also installed with the VOR.
 - VORTAC – When military tactical air navigations (TACAN) equipment is installed with a VOR
 - DME is always an integral part of a VORTAC.
- What are the different service volumes?



- What is the MON?
 - The VOR MON program is optimizing the VOR network in the contiguous United States (CONUS) to provide a backup conventional service in the event of an unplanned Global Positioning System (GPS) outage.
 - In order to establish the VOR MON, new VOR Standard Service Volumes (SSVs) are being implemented to provide VOR signal coverage starting at 5,000' Above Ground Level (AGL).
 - During an unplanned GPS outage, aircraft will be able to continue VOR-to-VOR navigation or proceed to a MON airport and conduct a VOR, Localizer LOC), or Instrument Landing System (ILS) approach.
 - 306 VORs that do not meet the VOR MON criteria are planned for discontinuance.
- How often is the inspection required?
 - Every 30 days for IFR flight.
- What are the different ways of testing the VOR?
 - VOR checkpoint (+/- 4°):

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- Located near a taxiway, ramp, or runup area. Follow instructions on the sign.
- VOT (approved VOR test signal located at the airport) (+/- 4 degrees):
 - Not all airports have one. Check chart supplement.
 - Tune VOR to VOT.
 - Set the course to 360 degrees, and the track indicator should be centered.
 - You should read FROM.
 - Next, set the course to 180.
 - And you should read TO.
- Dual VOR check (+/- 4°):
 - Easiest check as long as you have 2 VOR receivers.
 - Tune both NAV radios to the same VOR facility.
 - Center the needle to a TO indication.
- Airborne Check (+/- 6°):
 - VOR check in flight.
 - Select a VOR radial along the centerline of a VOR airway.
 - Select a ground point along the selected radial (preferably more than 20 nm) from the VOR facility.
 - Fly over the point and note the VOR bearing indicated by the receiver when over the point.
- What must be recorded?
 - DATE, PLACE, ERROR, and SIGNATURE in the aircraft log (DEPS).
- What are the limitations of the VOR?
 - Line of sight:
 - Obstacles, terrain, and slope of the earth can interfere.
 - Zone of ambiguity:
 - Occurs when perpendicular to a radial. The plane gets confused about whether it's TO or FROM.
 - Reverse sensing:
 - If the VOR is set to a TO indication, but you are actually coming FROM the station, you will have to turn opposite the needle.
 - Cone of confusion:
 - When the airplane is directly over the VOR, the VOR receiver will not be able to determine which radial the airplane is on.
 - If a VOR station is undergoing maintenance:
 - Either no morse code or test morse code (- -).
 - If the VOR function is not working, but the DME function is:
 - Single code ID every 30 seconds.
- DME:
 - How does the DME Work?
 - Airborne DME unit transmits an interrogation signal that the ground facility

receives and replies to the interrogation.

- The time passed is used by the airborne unit to calculate the slant range distance from the aircraft to the station $\text{Distance} = \text{Speed} \times \text{Time}$.
- Do we have a DME?
 - There is **not** a traditional DME receiver in the Avidyne. The G430 GPS substitutes for DME.
- What are the limitations of the DME?
 - Line of sight restrictions.
 - Maximum of 100 airplanes.
 - Slant Range Error:
 - For example, an aircraft directly above the DME station at 6,076 ft altitude would still show 1.0 NM on the DME readout
 - The aircraft is technically a mile away, just a mile straight up
 - Slant range error is most pronounced at high altitudes when close to the DME station.
- ILS:
 - Localizer:
 - Have the student explain/draw the function and service volume of the localizer.
 - Guidance is given from 18 nm from the antenna up to 4,500' above antenna elevation.
 - First 10 nm → 35 degrees on each side of the centerline.
 - Remaining 8 nm → 10 degrees on each side of the centerline.
 - 0 and 150 Hz signals are sent by the antenna and used by the receiver to determine the position on the localizer.
 - Course width: 3-6 degrees (usually 5) and full-scale deflection: 2.5 degrees.
 - What are the frequencies?
 - 108.1 and 111.95 MHz (odd tenths only).
 - What is the purpose of a localizer?
 - Provides horizontal guidance along the centerline of the runway.
 - Glideslope:
 - Have the student explain/draw the function and service volume of the glideslope.
 - Vertical position is determined by the aircraft using the 90 and 150Hz signals sent by the glideslope antenna.
 - Equipment is housed in a building approx. 750-1250' down from the approach end of the runway and 400-600' to one side of the centerline.
 - Range: 10 NM.
 - Width: 1.4 degrees and full-scale deflection is 0.7 degrees.
 - What are the frequencies?
 - 329.3 to 335 MHz (automatically tuned with the localizer).
 - What is the purpose of a glideslope?

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- Provides vertical guidance toward the runway touchdown point.
- What is a false glideslope?
 - Objects on the ground may reflect glideslope signals, leading to a glideslope indication much higher than where it should actually be located.
- Marker beacons:
 - What are marker beacons used for?
 - Provide range information along the approach path.
 - Have the student identify each beacon, its color, and morse code.
 - Outer marker (OM):
 - Blue and (- - -).
 - Located 4-7 miles out and indicates the position at which the aircraft intercepts the glideslope (at the appropriate altitude).
 - Middle marker (MM):
 - Amber and (- . -).
 - Located 3,500 ft from the landing threshold and indicates the point where the glideslope meets the decision height (usually 200ft above touchdown zone elevation).
 - Inner marker (IM):
 - White and (. . .).
 - Located between the middle marker and landing threshold. It indicates the point where the glideslope meets the decision height on a CATII ILS.
 - Back course marker (not part of the ILS system):
 - White and (. . .).
 - Indicates the final approach fix on a back-course approach.
- Approach lights:
 - What are approach lights used for?
 - Assist in the transition from instrument to visual flight.
 - What are the different types?
 - ALSF, SSALR, MALSR, REL, MALSF, ODALS, and VASIs.
- GPS:
 - How does a GPS receiver determine the aircraft position?
 - Time-lapse between transmission broadcast from atomic clocks on satellite to the time it was received by the aircraft transceiver is translated into distance. Using the intersection of spheres from multiple satellites, the aircraft's position is found.
 - How many satellites are required to define a fix?
 - At least 3 satellites are required for 2D position (latitude and longitude).
 - At least 4 satellites are required for 3D position (latitude, longitude, and altitude).
 - What are the 3 components of the GPS system?
 - Space:
 - Satellite constellation.

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- Composed of a constellation of 32 satellites approx. 11,000 NM above the earth.
- Must have 5 satellites in view at all times.
- User:
 - Antennas and receivers on the aircraft.
 - Must be certified for the type of use (IFR, VFR, Backup).
 - Verifies the signal's integrity using Receiver Autonomous Integrity Monitoring (RAIM).
- Control:
 - Ground-based monitoring system.
 - Master control station, 5 monitoring stations, and 3 ground antennas.
 - Ensures integrity of the satellite signal.
 - Corrects for distortion in satellite signal passing through the atmosphere.
- What are the sensitivities of the different GPS modes?
 - > 30 miles away:
 - Enroute Mode +/- 5NM (Non-WAAS) +/- 2NM (WAAS).
 - Switches to Terminal mode within 30 NM of the airport reference point.
 - < 30 miles away:
 - Terminal Mode +/- 1NM (Non-WAAS and WAAS).
 - Switches to Approach mode when 2 NM from the FAF.
 - 2 miles from FAF:
 - Approach Mode +/- 0.3NM (Non-WAAS and WAAS).
 - MAWP (Missed approach waypoint)
 - Suspend Mode +/- 1 NM
- How do we know if a GPS unit can be used for IFR operations?
 - GPS may be used as a primary means of navigation if:
 - The receiver is certified for the type of use.
 - TSO C129 or C196: certifies GPS for non-WAAS.
 - TSO C145 and C146: certifies GPS for WAAS.
 - The database is current.
 - RAIM signals can be read.
- RAIM:
 - What is the purpose of RAIM?
 - Function that monitors the integrity of the satellite signals.
 - What are the two modes of RAIM?
 - Fault Detection:
 - Requires a minimum of 5 satellites or 4 with baro-aiding altimeter.
 - Detects the presence of a positioning failure.
 - Fault Exclusion:
 - Requires 6 satellites or 5 with a baro-aiding altimeter.

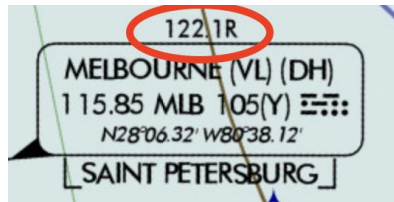
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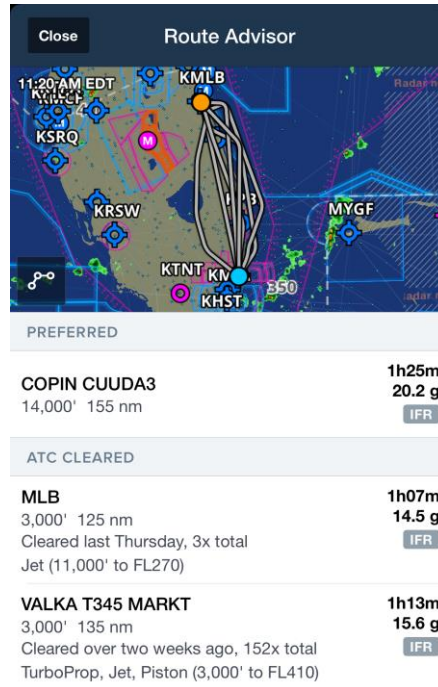
- Eliminates a corrupt satellite.
 - Upon detection, proper fault exclusion determines and excludes the source of the failure, thereby allowing GNSS navigation to continue without interruption.
- Do you need to check for RAIM prior to flying IFR?
 - Yes, a RAIM check is required before the flight. Otherwise, backup equipment must be used that is not GPS based.
- Is RAIM necessary with WAAS equipment?
 - No, WAAS can substitute for RAIM.
- **WAAS:**
 - What is the purpose of WAAS?
 - Improves the accuracy, integrity, and availability of GPS signals.
 - Allows for GPS approaches with vertical guidance (LPV or LNAV/VNAV).
 - How does WAAS work?
 - WAAS Reference stations on the ground detect errors in GPS signals.
 - The errors from the reference stations are sent to WAAS master stations and corrected.
 - Corrected signals are sent to a ground uplink station.
 - The ground uplink station sends signals to geostationary satellites.
 - Geostationary Satellites send the corrected signal to the aircraft's receiver.
 - How do you know if WAAS is on/available?
 - AUX page, SBAS Selection, WAAS is set to On.
 - Is backup equipment needed with WAAS?
 - No RAIM check is required, but you need to check WAAS NOTAMS.
- **GBAS:**
 - What is the purpose of GBAS?
 - Increases the accuracy of GPS and enhances the capabilities of the aircraft for approaches (GLS).
 - GBAS works similar to WAAS but is used just in the vicinity of an airport.
 - How does GBAS work?
 - 4 Reference stations located around an airport receive signals from GPS satellites.
 - The reference stations find the error in the satellite transmissions and send the computed information to the ground-based Facility.
 - Ground-based Facility corrects the information and, through a VHF data broadcast, sends the augmented signal to the aircraft where it can now be used for approaches with minimums as low as 200 feet AGL.

Cross-Country Flight Planning

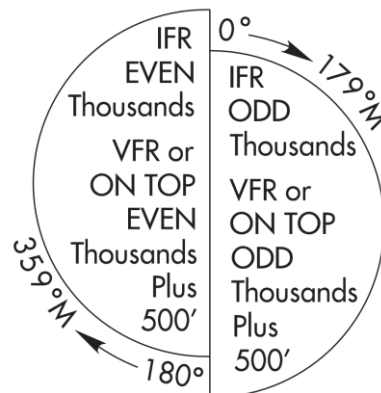
- What preflight action is required before an IFR cross-country flight?
 - 91.103 Preflight action - IFR flight or away from the vicinity of an airport:
 - NOTAMs.

- Weather.
- Known delays.
- Runway lengths.
- Alternates if needed.
- Fuel requirements.
- Takeoff/landing data.
- How to file an IFR flight plan?
 - Flight Service Station (FSS)
 - Phone: 1-800-WX-BRIEF
 - Radio:
 - Frequencies are shown in chart supplement or on the LE chart.
 - RCO/GCO - check chart supplement.
 - Online: 1800wxbrief.com
 - EFB
 - Foreflight: Flights tab
- How to determine an IFR cross-country route?
 - ATC preferred routes
 - Located in the chart supplement
 - Check requirements of airways, SIDs, and STARs to determine if they can be flown
 - EFB
 - Some flight planning apps/services can create routes as well as show how often they're cleared, the aircraft and altitudes the route is designed for, etc.





- Create your own route
 - Use airways as much as possible
 - Check requirements of airways, SIDs, and STARs to determine if they can be flown
- Which altitudes are suitable for IFR cross-county flying?



- What are the fuel requirements for an IFR flight? FAA and FIT?
 - 91.167 - Fuel requirements for flight in IFR conditions:
 - Departure to destination airport
 - +
 - Most distant alternate
 - +
 - 45 minutes at normal cruise
 - FIT FOM 3.4.7 Fuel Reserves - Usable Fuel:
 - One hour of reserve at all times.

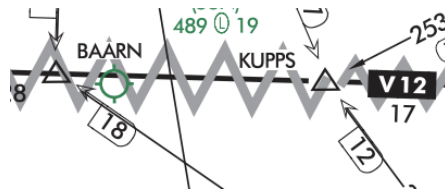
- **Alternate airport requirements**

- When is an alternate required (91.169)?
 - An alternate is NOT required when the DESTINATION airport:
 - Has an instrument approach procedure
 - AND
 - Weather reports and forecasts indicate that from 1 hour before to 1 hour after ETA, the ceiling will be at least 2,000' AGL and the visibility will be at least 3SM (1,2,3 rule).
 - What are the IFR alternate airport minimums?
 - Standard Alternate Minima - 91.169:
 - Precision: 600 ft ceiling and 2 statute miles.
 - Non-precision: 800 ft ceiling and 2 statute miles.
 - No instrument approach available: descent from the MEA and landing must be conducted under basic VFR.
 - What does the  designation on the approach chart indicate?
 - The approach procedure has non-standard minimums (for aircraft other than helicopters) or restrictions (for all users) for its use as an alternate.

 ALTERNATE MINS	
24081	
NAME	ALTERNATE MINIMUMS
MELBOURNE, FL	
MELBOURNE ORLANDO	
INTL (MLB).....	ILS or LOC Rwy 9R ¹²
	LOC BC Rwy 27L ¹³
	RNAV (GPS) Rwy 9L ³
	RNAV (GPS) Rwy 9R ³
	RNAV (GPS) Rwy 27L ³
	RNAV (GPS) Rwy 27R ³
	VOR Rwy 9R ³
	VOR Rwy 27L ³
¹ NA when control tower closed.	
² LOC, Category D, 900-2¾.	
³ Category D, 900-2¾.	

- What does an  NA designation on the approach chart indicate?
 - Pilots may not use that approach as an alternate due to unmonitored facility, absence of weather reporting service, or lack of adequate navigation coverage.
 - What are some other considerations to make when selecting an alternate?
 - Ensure that fuel requirements can be complied with (91.167)
 - Make sure that the airport has approaches that are approved to be used as an alternate (does not have the  NA).
 - Check for any non-standard alternate minima.
 - What if your aircraft is not equipped with WAAS?
 - Either the destination or alternate must be equipped with a ground-based instrument approach system (AIM 1-1-17).

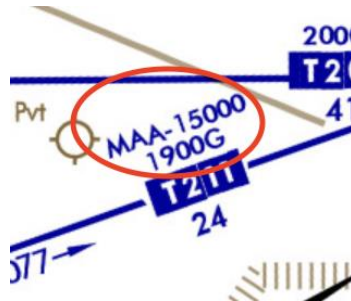
- Consider runway length, lighting, services available, etc.
- Quiz the student on the low-altitude enroute chart.
 - How high do they go?
 - Up to 18,000ft MSL.
 - How often are they updated?
 - Every 56 days.
 - Airports:
 - Ask the student to read the information text of an airport.
 - What do the different colors of the information text of airports indicate?
 - Airports in **green** have instrument approach procedures.
 - Airports in **blue** have high-altitude military approach procedures.
 - Airports in **brown** do not have a published approach procedure.
 - Airspace:
 - What do the blue shaded regions indicate?
 - Class B or C airspace.
 - What do the blue or brown dashed outlined regions indicate?
 - Special use airspace.
 - Airways:
 - What is the purpose of an airway?
 - Efficiency and traffic separation.
 - How wide are they?
 - Usually 8nm wide (4nm each side).
 - What are the types of airways?
 - Victor airways are based on VORs and are depicted in **Black**.
 - Tango airways are based on GPS waypoints and are depicted in **Blue**.
 - Bravo or Alpha airways are based on NDBs and are depicted in **Amber**.
 - What does it mean when an airway contains a combination of colors?
 - More than one type of navigation can be used for a segment.
 - Why are some airways depicted as shown?
 - The segment is unusable.



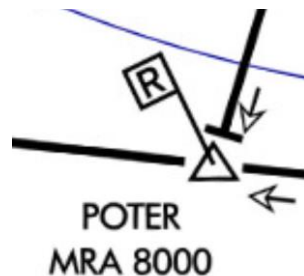
- IFR altitudes: find an example to ask the student about each of the following:
 - What are the minimum IFR altitudes?
 - Except for takeoff or landing... no person may operate an aircraft under IFR below:
 - Minimum altitudes prescribed for the flown segment, or if none:

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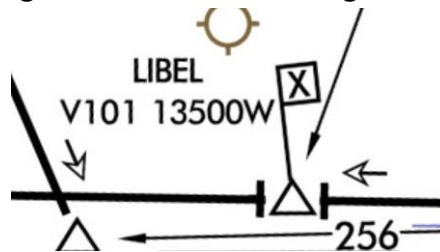
- Non-mountainous areas: +1,000ft
- Mountainous areas: +2,000ft
- What is a Maximum Authorized Altitude (MAA)?
 - Maximum usable altitude or flight level for an airspace structure or route segment.



- What is a Minimum Reception Altitude (MRA)?
 - Minimum altitude the navigation signal for the route and for off-course NAVAID facilities that determine a fix.

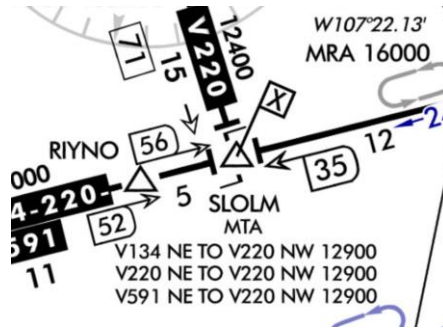


- What is a Minimum Crossing Altitude (MCA)?
 - Lowest altitude at certain fixes at which the aircraft must cross when proceeding in the direction of a higher minimum enroute IFR altitude.



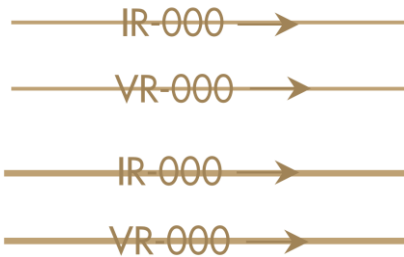
- What is a Minimum Turning Altitude (MTA)?
 - Lowest altitude at certain fixes where aircraft are expected to turn from one airway onto another. Guarantees obstacle protection should the aircraft fly over a fix (overshoot) during the turn.

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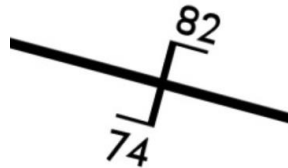


○ Miscellaneous symbology:

- What are Military Training Routes (MTRs)? Why are some lines thin and some are thick?
 - Depicted as a solid brown line, indicating low-altitude, high-speed training.
 - All IR (IFR) and VR (VFR) MTRs are shown except for VRs at or below 1,500ft AGL.
 - Thin lines: 5NM or less both sides of centerline.
 - Thick lines: greater than 5NM either or both sides of centerline.

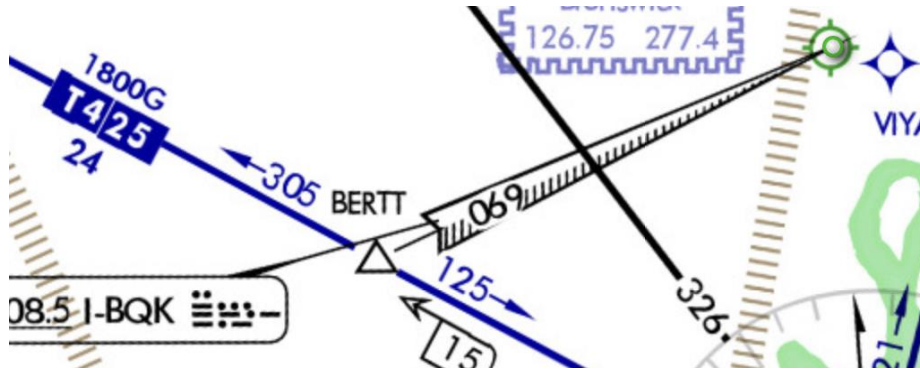


- What is a VOR changeover point?
 - Indicates the location along an airway (depicted as DME distances along an airway) at which the aircraft must change VOR frequency.
 - What if one is not depicted?
 - Change VOR frequency at the midpoint or bend of an airway.



- Why are some localizers depicted on the low enroute chart? Why not all?
 - Localizer course with additional navigation function indicates a localizer which may also be used to identify a fix on an airway.
 - In this example, the I-BQK localizer may also be used to identify BERTT, a fix on T425:

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- What does an MON designation indicate?
 - Indicates the airport has at least one instrument approach procedure not requiring the use of GPS.

MON
Miami Exec
(TMR)

- What are the required position reports under IFR (MARVELOUSVFR500)? Which ones are only required in non-radar environments?
 - Missed approach
 - Airspeed ± 10 kts / 5% change of filed TAS (whichever is greater)
 - Reaching a holding fix (report time & altitude)
 - VFR on top when an altitude change will be made.
 - ETA changed ± 2 min, or ± 3 min in North Atlantic (NAT) *
 - Leaving a holding fix/point
 - Outer marker (or fix used in lieu of it) *
 - Unforecasted weather
 - Safety of flight
 - Vacating an altitude / FL
 - Final Approach fix *
 - Radio/Nav/approach equipment failure
 - Compulsory reporting points ▲ *
 - 500 - unable to climb/descend at 500 fpm

* Only required in non-radar environments (acronym: COFE)

Air Traffic Control Clearances

- When is a clearance required? Can we fly into IMC in uncontrolled airspace without a clearance?
 - According to 14 CFR 91.173, no person may operate an aircraft under IFR in controlled airspace unless an IFR flight plan has been filed and a clearance has been received from ATC.
 - Although flying into IMC in uncontrolled airspace without a clearance is

technically legal, it has been found in the past to be a violation of **91.13 - Careless or reckless operation**.

- How to obtain a clearance? Can you depart IFR after the clearance void time?
 - Towered airport: Contact clearance delivery or ground frequency. The ATIS will sometimes specify a frequency.
 - Non-towered airport: Consult chart supplement / AFD for frequency or ARTCC phone number.
 - Depart VFR and contact ATC.
 - Contact ATC on the ground. Expect a *Clearance Void Time* or *EDCT (Expect Departure Clearance Time)*.
 - Clearance void time: The latest time at which one may depart IFR. A pilot who does not depart prior to the clearance void time must advise ATC as soon as possible of their intentions.
 - Ground communication outlet (GCO): on the ground, click mic 6 times to contact FSS & 4 times to contact ATC.
 - Elements of an IFR clearance:
 - C: Clearance Limit
 - R: Route
 - A: Altitude
 - F: Frequency
 - T: Transponder

TECH FLIGHT 636, Cleared to the Sarasota International Airport as filed, after departure maintain 2000ft, expect 6000ft 10 minutes after departure, departure frequency is 132.65, squawk 0812.

- What is a cruise clearance?
 - The term “cruise” may be used instead of “MAINTAIN” to assign a block of airspace to a pilot from the minimum IFR altitude up to and including the altitude specified in the cruise clearance.
 - The pilot may level off at any intermediate altitude within this block of airspace. Climb/descent within the block is to be made at the discretion of the pilot. However, once the pilot starts descent and verbally reports leaving an altitude in the block, the pilot may not return to that altitude.

Compliance with Departure, Enroute, and Arrival Procedures and Clearances

- Takeoff Minimums:
 - What are the Standard IFR Takeoff Minimums?
 - Part 121, 125, 129, or 135:
 - Aircraft with 2 engines or less: 1 statute mile visibility.
 - Aircraft with more than 2 engines: 1/2 statute mile visibility.

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- Helicopters: 1/2 statute mile visibility.
- No pilot may takeoff under IFR from a civil airport having published obstacle departure procedures (ODPs) for the takeoff runway to be used, unless the pilot uses such ODPs or an alternative procedure or route assigned by air traffic control.
 - PART 91: NO T/O MINS.
- What are the FIT Takeoff Minimums?
 - The ceiling and visibility (both reported and forecast) must be sufficient to complete an IFR approach at the departure airport.
 - An alternate airport must be available within a one-hour flight from the departure airport if the weather is less than VFR at that departure airport.
- **Departure procedures:**
 - What is the purpose of a departure procedure?
 - Provides a transition from the airport to the enroute structure while ensuring obstruction clearance.
 - Are we required to file a departure procedure?
 - Pilots operating under Part 91 are encouraged to file and fly a departure procedure when one is available.
 - What are the criteria required to provide obstruction clearance on a departure procedure?
 1. Climb at 200ft/NM until told to level off or upon reaching the minimum IFR altitude.
 2. Cross the departure end of the runway (DER) after reaching 35' AGL.
 3. Make the initial turn only after reaching 400' AGL.
 - How to convert 200 ft/NM into fpm?
 - What are the types of Departure Procedures?
 - Obstacle Departure Procedure (ODP)
 - Where do we find them? What are the types?
 - Textual ODPs will be listed under DEPARTURE PROCEDURE in the IFR Takeoff Minimums, Departure Procedures, and Diverse Vector Area (Radar Vectors)  section of the TPP.
 - Graphical ODPs will include (OBSTACLE) in the title and will be found in the Terminal Charts section of the TPP.
 - What do they provide?
 - Obstruction clearance from the terminal area to the enroute structure ONLY.
 - Are they mandatory to follow?
 - No, but recommended (e.g. night, IMC, terrain).
 - Do they require an ATC clearance?
 - No.

- Standard Instrument Departure (SID)
 - Where do we find them? What are the types?
 - Can be found in the Terminal Charts section of the TPP, just after the approach procedures for a particular airport.
 - Graphical only.
 - What do they provide?
 - Obstruction clearance from the terminal area to the enroute structure.
 - Pilot/controller workload reduction.
 - Do they require an ATC clearance?
 - Yes.
 - Ask the student to find a SID (for example, the DORRL TWO @ KMIA)
 - Ask the student to brief the SID. Make sure they identify whether their aircraft is allowed/has the performance to comply with the instructions.
 - Quiz the student on SID symbology.
- Diverse Vector Areas
 - What is a Diverse Vector Area?
 - An area in which ATC may provide radar vectors until the aircraft reaches a safe altitude.
 - Where do we find them?
 - The Takeoff Minimums, Obstacle Departure Procedures, and Diverse Vector Area  section of the TPP.
- Visual Climb Over Airport (VCOA)
 - What is a VCOA?
 - A departure option for an IFR aircraft involving a climb while remaining over the field, operating in VMC equal to or greater than the specified visibility and ceiling to altitude where the aircraft can be in IMC and proceed via radar vectors or a fix mentioned in the departure procedure.
 - Where can we find them?
 - The Takeoff Minimums, Obstacle Departure Procedures, and Diverse Vector Area section of the TPP.
 - What are Takeoff Obstacle Notes?
 - Low close-in obstacles are those obstacles that are located within 1 NM of the DER and penetrate the 40:1 Obstacle Clearance Surface (OCS).
 - Low close-in obstacles are noted in the IFR Takeoff Minimums, Departure Procedures, and Diverse Vector Area (Radar Vectors) section .
 - **Arrival procedures:**
 - Standard Terminal Arrival Routes (STARs):
 - What is a STAR?

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- These arrival routes simplify clearance delivery procedures for arriving aircraft, specifically for airports that have a high density of traffic.
 - Where are they located?
 - STARs are located towards the front of the TPP in the Standard Terminal Arrival Charts section.
 - How are they depicted?
 - Textual and graphical.
 - Do we request them?
 - ATC may issue a STAR.
 - To accept this clearance, a pilot must at least have the textual description.
 - It is the pilot's responsibility to accept or deny the issuance of a STAR.
 - What if we do not want ATC to assign a STAR?
 - A pilot may file an IFR flight plan and put "NO STAR" in the remarks.
 - If a STAR is accepted, it requires mandatory compliance.
 - Ask the student to find a SID (for example, the PRICY THREE @ KMLB)
 - Ask the student to brief the SID. Make sure they can identify whether their aircraft can comply with the instructions, such as speed and altitude restrictions
- **Approaches**
 - What is a precision approach?
 - A standard instrument approach procedure in which an electronic glideslope or other type of glidepath is provided.
 - What are the types of precision approaches?
 - ILS
 - GLS
 - PAR: The controller issues guidance instructions to the pilot based on the aircraft's position in relation to the final approach course (azimuth), the glidepath (elevation), and the distance (range) from the touchdown point on the runway as displayed on the radar scope.
 - What is a non-precision approach?
 - A standard instrument approach procedure in which no electronic glideslope is provided.
 - What are the types of non-precision approaches?
 - VOR
 - TACAN
 - NDB
 - LOC
 - ASR
 - LDA
 - SDF
 - What is an approach with vertical guidance?

- A term used to describe RNAV approach procedures that provide lateral and vertical guidance but do not meet the requirements to be considered a precision approach.
- What are the types of APV approaches?
 - LPV: Localizer Performance with Vertical guidance - the CDI needle becomes more sensitive the closer to the runway, like a localizer. Requires WAAS.
 - LNAV/VNAV: Lateral and Vertical Navigation - the CDI needle does not increase in sensitivity. Requires WAAS.
 - LNAV: Lateral Navigation only - used when no Vertical Navigation exists (e.g. PFD failure).
- What is a Contact approach?
 - An approach wherein an aircraft on an IFR flight plan, having an air traffic control authorization, operating clear of clouds with at least 1 mile flight visibility and a reasonable expectation of continuing to the destination airport in those conditions, may deviate from the instrument approach procedure and proceed to the destination airport by visual reference to the surface.
- Can ATC assign a Contact approach?
 - No, this approach will only be authorized when requested by the pilot.
- What does the weather need to be for a Contact approach?
 - The reported ground visibility at the destination airport is at least 1 statute mile.
- What is a Visual approach?
 - An approach conducted on an instrument flight rules (IFR) flight plan which authorizes the pilot to proceed visually and clear of clouds to the airport.
- What must the pilot keep in sight to conduct a Visual approach?
 - The pilot must, at all times, have either the airport or the preceding aircraft in sight.
- Can ATC assign a Visual approach?
 - Yes, this approach can be authorized and under the control of the appropriate air traffic control facility.
- What does the weather need to be for a Visual approach?
 - Ceiling at or above 1,000 feet, and visibility of 3 miles or greater.
- What is a No-gyro approach?
 - A radar approach/vector provided in case of a malfunctioning gyro-compass or directional gyro.
 - Instead of providing the pilot with headings to be flown, the controller observes the radar track and issues control instructions “turn right/left” or “stop turn” as appropriate.
- Ask the student to brief an approach plate. Make sure their briefing includes the following information:
 - Briefing strip - contains general information about the approach.
 - Approach name
 - Airport and airport code

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- Approach frequency
- Final approach course (note that this may not always be what we twist the CDI to, such as on a LOC BC)
- Runway landing distance (Rwy Ldg)
- Touchdown zone elevation (TDZE)
- Airport elevation (Apt Elev)
- General notes about the approach, such as equipment requirements or anything that is nonstandard (includes takeoff notes and nonstandard alternate minimums).
- Approach lighting: depicted as a name, code, and diagram. Refer to the legend (around page I2) for further information.
- Missed approach instructions: textual description
- Airport and approach control frequencies
- Plan view - a top-down view which mainly depicts the lateral track of the approach.
 - Final approach course
 - Navaid track (localizer/radial/etc.)
 - IAF/IF/FAF locations and means of identification
 - Missed approach instructions: graphical view
 - MSA: Minimum Safety Altitude
 - Provides at least 1,000 feet of obstacle clearance within a 25-mile radius of the navigation facility, waypoint, or airport reference point upon which the MSA is predicated. MSAs are for emergency use only and do not necessarily assure acceptable navigational signal coverage.
- Profile view - a “side” view which mainly depicts the vertical track of the approach.
 - IAF/IF/FAF of the approach and their distances to the runway and/or navaid
 - Stepdown fixes
 - A fix permitting additional descent within a segment of an instrument approach procedure by identifying a point at which a controlling obstacle has been safely overflown.
 - Depicted as solid lines above and/or below crossing altitudes. 1600
 - Glideslope depiction
 - Glideslope angle and Threshold Crossing Height (TCH)
 - Glideslope intercept point and altitude
 - VDP (explained below)
 - Missed approach point
 - Missed approach instructions: graphical overview
 - Minimums (explained below)
- Airport diagram
 - Airport elevation
 - Touchdown zone elevation

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- Final approach course is depicted as an arrow. Useful for determining direction to turn when circling.
 - Approach lighting
 - Airport lighting
 - Runway layout, lengths, and widths
 - **D** symbol indicates Runway Declared Distance information is found in the Chart Supplement.
- Segments of an approach
 - What are the segments of an approach?
 - Feeder routes
 - A route depicted on IAP charts to designate routes for aircraft to proceed from the en route structure to the IAF.
 - Depicted as thin lines.
 - DME arcs
 - Some approaches include DME arcs as part of the approach procedure itself.
 - The aircraft may need to descend during the arc (ex: VOR Rwy 15 @ KMTN).
 - Course reversal
 - Some approach procedures do not permit straight-in approaches unless pilots are being radar vectored. In these situations, pilots are required to complete a procedure turn (PT) or other course reversal, generally within 10 NM of the PT fix, to establish the aircraft inbound on the intermediate or final approach segment.
 - When is a procedure turn not required? (SHARPTT)
 - Straight-in approach clearance
 - Holding in lieu of a procedure turn
 - DME Arc
 - Radar vectors to final
 - No PT depicted on chart
 - Timed approach from a hold fix
 - Teardrop course reversal
 - Initial approach segment
 - The segment between the initial approach fix (IAF) and the intermediate fix (IF) or the point where the aircraft is established on the intermediate course or final approach course.
 - Intermediate approach segment
 - The segment between the intermediate fix or point (IF) and the final approach fix (FAF).
 - Final approach segment
 - The segment between the final approach fix or point (FAF) and the runway, airport, or missed approach point (MAP).

- Missed approach segment
 - The segment between the missed approach point (MAP) or the point of arrival at decision height and the missed approach fix at the prescribed altitude.
- VDP
 - What is a VDP?
 - Identified by the symbol () , is a defined point on the final approach course of a non-precision straight-in approach procedure from which a stabilized visual descent from the MDA to the runway touchdown point may be commenced. The VDP will be identified by DME or RNAV along-track distance to the MAP.
 - Can we descend below the MDA before the VDP?
 - The pilot should not descend below the MDA prior to reaching the VDP.
 - What if a VDP is not published?
 - To calculate VDP when not published = $MDA (AGL)/300$, this ensures a three-degree slope to the runway
- Minimums
 - What is the difference between a DA and an MDA?
 - DA: a specified altitude in the precision approach at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.
 - MDA: the lowest altitude, expressed in feet MSL, to which descent is authorized on final approach or during circle-to land maneuvering in execution of a standard instrument approach procedure (SIAP) where no electronic glideslope is provided.
 - Quiz the student on each of the values (DA, DH, MDA, visibility, etc.). in the landing minima section on an approach plate.
 - What happens if the approach lighting is inoperative?
 - The required visibility will increase. Refer to the Inoperative Components or Visual Aids page in the front of the TPP. Make sure the student can correctly use the tables.
 - What happens if the approach plate requires RVR and the equipment is inoperative?
 - RVR will need to be converted to SM using the Comparable Values of RVR and Visibility table in the Terms/Landing Minima Data pages.
 - When can we descend below the MDA/DA?
 1. The aircraft is continuously in a position from which a descent to a landing on the intended runway can be made at a normal rate of descent using normal maneuvers, and for operations conducted under part 121 or part 135 unless that descent rate will allow touchdown to occur within the touchdown zone of the runway of intended landing;
 2. The flight visibility is not less than the visibility prescribed in the standard instrument approach being used.
 3. Except for a Category II or Category III approach where any necessary visual reference requirements are specified by the Administrator, at least one of the

following visual references for the intended runway is distinctly visible and identifiable to the pilot:

- The approach light system, except that the pilot may not descend below 100 feet above the touchdown zone elevation using the approach lights as a reference unless the red terminating bars or the red side row bars are also distinctly visible and identifiable.
 - The threshold.
 - The threshold markings.
 - The threshold lights.
 - The runway end identifier lights.
 - The visual glideslope indicator.
 - The touchdown zone or touchdown zone markings.
 - The touchdown zone lights.
 - The runway or runway markings.
 - The runway lights.
- Missed approach
 - When do we execute our missed approach?
 - If we are using a glideslope, execute the missed approach upon reaching the DA.
 - If we are not using a glideslope, hold the Minimum Descent Altitude (MDA) until reaching the Missed Approach Point (MAP).
 - When do we execute our missed approach at an airport whose MAP is not published? (Ex: ILS 36 @ KTIKX?)
 - Use the “FAF to MAP” box shown at the bottom of the airport diagram on the approach plate to calculate the MAP.
 - Show the student examples of several different approaches and make sure they can identify when to go execute a missed approach on all of them.
- Circling approach
 - What is a circling approach? When might we use one?
 - A maneuver initiated by the pilot to align the aircraft with a runway for landing when a straight-in landing from an instrument approach is not possible or is not desirable. (Ex: weather, runway length, lighting, etc.)
 - At tower-controlled airports, this maneuver is made only after ATC authorization has been obtained and the pilot has established required visual reference to the airport.
 - What does the  symbol indicate?
 - Expanded Circling Approach Maneuvering Airspace Radius
 - Table is found in the Terms/Landing Minima Data section of the TPP.
 - What do approaches that end in letters at the beginning of the alphabet (RNAV-A, LOC/DME-E, etc.) indicate? Why?
 - Approach is circling-only. Circling-only approaches are published for the following

criteria:

1. The final approach course alignment with the runway centerline exceeds 30°.
2. The descent gradient is greater than 400 ft/NM from the FAF to the threshold crossing height (TCH).
3. A runway is not clearly defined on the airfield.

Holding Procedures

- Holding speeds and altitudes:
 - What are the maximum holding airspeeds?
 - Below 6,000 ft. MSL: 200 KIAS.
 - 6,001 ft. MSL to 14,000 ft. MSL: 230 KIAS.
 - Above 14,000 ft. MSL: 265 KIAS.
 - What are the standard holding leg lengths?
 - At or below 14,000 feet MSL: 1:00
 - Above 14,000 feet MSL: 1:30

Loss of Communications

- What should you do if you suspect you have lost communications?
 1. Troubleshoot
 - Return to last assigned frequency
 - Check volume of COM1/COM2 and headset
 - Unplug/replug headset, try different jacks
 - Reset COM1/COM2 circuit breaker
 - Test Squelch function
 - If you are familiar with the airport and have access to the control tower phone number, depending on if you have cellular network, call up the tower. Instructors, share MLB tower's phone number with your students: (321) 768-1524.
 2. Squawk 7600
 3. Determine if VMC (Visual Meteorological Conditions) or IMC (Instrument Meteorological Conditions):
 - If VMC, land as soon as practicable.
 - If IMC, determine a route and altitude to fly to the destination:

Route - in order:

Assigned
Vectored
Expected
Filed

Altitude - highest of:

Minimum IFR Altitude
Expected
Assigned

- If the clearance limit is a fix from which an approach begins, begin the approach at EFC (or ETA, if none exists).

- If the clearance limit is not a fix from which an approach begins, leave clearance limit at EFC (or ETA, if none exists) and proceed to a fix from which an approach begins.
- Give the student several scenarios to test their knowledge of the procedure. For example:
 - Their clearance limit is the destination airport. They have lost comms right after departure, while flying runway heading.
 - This is a dead vector. Fly runway heading until reaching at least 400' AGL, then turn to their expected or filed route. If they reach the destination airport prior to the ETA, hold over the destination airport at or above the MSA.
 - Their clearance limit is a fix on an airway. Their last instruction was to "fly heading ___ to join V___, climb and maintain 2,000, expect 4,000 in 10 minutes."
 - Fly the vector, then join the airway. Make sure the student correctly identifies the OROCA/MOCA for the vector and airway, then compares it to 4,000 to fly the highest. Upon reaching the fix, hold at that fix if before the EFC. At the EFC, fly direct to an IAF and begin the approach.
 - Their clearance limit is an IAF. They reach the IAF before the EFC.
 - Hold at the IAF until the EFC, then begin the approach. If there is a published hold as part of the approach, hold at the published altitude. If the hold is not part of the approach, hold at the MSA.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 331b.
- Assign Homework:
 - Review all instrument flight procedures in preparation for the flight.

Completion Standards

At the completion of this lesson, the student shall:

1. Display a complete understand of IFR procedures.
2. Demonstrate the necessary knowledge and skill to perform all IFR procedures at or above the proficiency level listed in the current FAA Instrument Rating (Airplane) Airman Certification Standards.